:- 12th BOARD HACKER JUGAAD NOTES :-

DIFFERENTIAL EQUATION

1. An equation containing independent variable, dependent variable and differential coefficients of dependent variables with respect to independent variable is called differential equation.

$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}$ → Differential Coefficient

$y \rightarrow$ dependent variable

$x \rightarrow$ independent variable

$$\boxed{\frac{d^2y}{dx^2} + \frac{y}{x} = 0} \rightarrow \text{D.E}$$

2. If all the differential coefficients are present with respect to independent variable then the differential equation is called ordinary differential equation. If they are present more than one independent variables then it is called partial differential equation.

eg $\frac{dy}{dx} + \frac{d^2y}{dx^2} + \frac{y^2}{x} = 0$

→ independent variable $= x$

→ ordinary D.E

$\underbrace{\frac{d^2y}{dx^2} + \frac{dy}{dz} + x = 0}_{\text{out of syllabus}}$ → independent variable $= x$ & z

→ Partial D.E.

3. Differential equation represent a family of curves which share some common characteristics.

eg → $y = mx$ → family of straight lines passes through origin.

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\Rightarrow \log y = \log x + \log c$$

$$\Rightarrow \log y = \log xc$$

$$\boxed{y = xc}$$

$y = xc$
$y = 2x$
$y = 5x$
$y = 8x$
$y = \sqrt{2}x$

} Straight lines passing through Origin.

Order and Degree

1. The order of highest Order differential Coefficient present in the equation is called order of differential equation.

$$\frac{d^2y}{dx^2} + \frac{d^3y}{dx^3} + \left(\frac{d^4y}{dx^4}\right)^3 + \frac{d^5y}{dx^5} = 0$$

Order = 5

2. The exponent of highest order differential Coefficient present in the equation is called degree of differential equation provided if all the differential coefficients are present in polynomial form.

Order of a differential equation is always defined while degree need not to be defined.

eg. $\left(\frac{d^3y}{dx^3}\right) + \left(\frac{d^2y}{dx^2}\right)^7 + \left(\frac{dy}{dx}\right)^{25} = 0$

Degree = 1
Order = 3

eg. $\left(\frac{d^5y}{dx^4}\right)^7 + \left(\frac{d^2y}{dx^2}\right)^{100} = 0$

Degree = 7
Order = 5

POLYNOMIAL FORM

→ Power of differential Coefficients is whole no.

eg. $\left(\frac{d^3y}{dx^3}\right)^1 + \left(\frac{d^2y}{dx^2}\right)^8 = \frac{x}{y}$

(In polynomial form)

eg. $x^7y^8 + \left(\frac{d^3y}{dx^3}\right)^8 + \sqrt{\frac{dy}{dx}} = 0$

(Not in polynomial form)

eg. $\frac{d^5y}{dx^5} + \left(\frac{d^2y}{dx^3}\right)^{2/3} + y = 0$

(Not in Polynomial form)

eg. $\left(\frac{d^3y}{dx^3}\right)^1 + \left(\frac{d^2y}{dx^2}\right)^7 + x^{3/2} = 0$

(In polynomial form)

eg. $\sin\left(\frac{dy}{dx}\right) = x^5$

(Not in Polynomial form)

eg. $\frac{d^2y}{dx^2} = \sin(x^5)$

(In polynomial form)

eg. $\tan^{-1}\left(\frac{d^2y}{dx^2}\right) = \frac{dy}{dx}$

(Not in Polynomial form)

eg. $\frac{dy}{dx} = e^{\frac{d^2y}{dx^2}}$

(not in polynomial form)

eg. $\log\left(\frac{dy}{dx}\right) = x^3$

(not in polynomial form)

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eg. $\sin\left(\frac{dy}{dx}\right) = x^5$

$\Rightarrow \frac{dy}{dx} = \sin^{-1}(x^5)$

→ Order = 1

Degree = 1

eg. $\sin\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$

$\frac{dy}{dx} = \sin^{-1}\left[\frac{d^2y}{dx^2}\right]$

order = 2

Degree = not defined.

$\cos\left(x + \frac{dy}{dx}\right) = x^3$

$x + \frac{dy}{dx} = \cos^{-1}(x^3)$

degree = 1

order = 1

eg. $\left(\frac{dy}{dx}\right)^{\frac{3}{2}} = \left(\frac{d^2y}{dx^2}\right)^4$

Squaring both side

$\left[\left(\frac{dy}{dx}\right)^{3/2}\right]^2 = \left[\left(\frac{d^2y}{dx^2}\right)^4\right]^2$

$\left(\frac{dy}{dx}\right)^3 = \left(\frac{d^2y}{dx^2}\right)^8$

order = 2

degree = 8

eg. $y'' = (y')^{\frac{1}{3}}$

Cube both side

$(y'')^3 = \left((y')^{\frac{1}{3}}\right)^3$

$(y'')^3 = (y')^1$

degree = 3

eg. $(y')^{\frac{2}{3}} = (y'')^{\frac{7}{2}}$

$\left((y')^{\frac{2}{3}}\right)^{\times 2} = \left((y'')^{\frac{7}{2}}\right)^{\times 3}$

$(y')^4 = (y'')^{21}$

order = 2

degree = 1

General and particular Solutions of a differential Equation

1. The family of curves obtained by the integrating the given differential equation, free from differential coefficients and containing the arbitrary constants are called general solutions.

$$\frac{dy}{dx} = \frac{y}{x} \rightarrow \text{D.E}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\boxed{\log|y| = \log|x| + c}$$

↓ General Solution ↓ Arbitrary Constant

General and particular solutions of a differential Equation

2. If the arbitrary constants present in general solution are substituted with some specific values then the obtained equation is called the particular solution.

$\log y = \log x + c$ ↘ General Solution

↓ Curve passes theory (2,1)

$x = 2$ & $y = 1$

$\log 1 = \log 2 + c$

$0 = \log 2 + c$

$\boxed{c = -\log 2}$

$\log|y| = \log|x| + c$

$\boxed{\log|y| = \log|x| - \log 2}$

↳ Particular solution

** No. of arbitrary constants in particular solutions = 0.

Methods of solving differential Equation

- ① VARIABLE SEPARABLE
- ② HOMOGENEOUS D.E
- ③ LINEAR D.E

Methods of Solving Differential Equation.

Variable Separable

Separate the variable & then integrate

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\log|y| = \log|x| + c$$

$$\Rightarrow \log|y| = \log|x| + \log|c|$$

$$\Rightarrow \log|y| = \log|x \times c|$$

$$\Rightarrow |y| = |xc|$$

$$\Rightarrow y = xc$$

Homogeneous Differential Equations

Any differential equation in which right hand side of the given equations is of the form of

$$\frac{dy}{dx} = F(x, y)$$

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right)$$

$$\frac{dy}{dx} = \frac{y^2+x^2}{xy}$$

① $x \to \lambda x$, $y \to \lambda y$

$$\frac{dy}{dx} = \frac{(\lambda y)^2 + (\lambda x)^2}{\lambda x \times \lambda y}$$

$$\frac{dy}{dx} = \frac{\lambda^2 y^2 + \lambda^2 x^2}{\lambda^2 xy}$$

$$\frac{dy}{dx} = \frac{x^2(y^2+x^2)}{x^2 xy} = \frac{y^2+x^2}{xy}$$

$$\frac{dy}{dx} = \frac{x^2+y^2}{xy}$$

② $$\frac{dy}{dx} = \frac{y^2+x^2}{xy}$$

y^2 → 2, x^2 → 2, $x'y'$ → 2

③ $$\frac{dy}{dx} = \frac{\frac{y^2}{x^2} + \frac{x^2}{x^2}}{\frac{xy}{x^2}}$$

$$\frac{dy}{dx} = \frac{\frac{y^2}{x^2} + 1}{y/x}$$

$$\frac{dy}{dx} = \frac{\left(\frac{y}{x}\right)^2 + 1}{\left(\frac{y}{x}\right)}$$

Linear differential Equation

1. A differential equation is of the form $\frac{dy}{dx} + Py = Q$, where P & Q are Continuous functions of x only, as known as a first order differential equation.

$$\boxed{\frac{dy}{dx} + Py = Q}$$

P & Q ⇒ Constant or Continuous function of variable x only

⇒ Coff of $\frac{dy}{dx} = +1$

⇒ P = Coff of y only

⇒ Q = should be present on RHS.

eg. ① $\frac{dy}{dx} + 2y = 5$; $P = 2$, $Q = 5$

② $\frac{dy}{dx} + (\sin^2 x)\,y = \sec^2 x$; $P = \sin^2 x$, $Q = \sec^2 x$

③ $\frac{dy}{dx} + (\sin x)\,y + 7\cos x + 5y = 0$

$\frac{dy}{dx} + y(\sin x + 5) = -7\cos x$

$P = \sin x + 5$

$Q = -7\cos x$

④ $x^2\frac{dy}{dx} + y\sin^2 x + (\cos^2 x)\,y = 0$

$\frac{x^2\frac{dy}{dx}}{x^2} + \frac{y(\sin^2 x + \cos^2 x)}{x^2} = \frac{0}{x^2}$

$\frac{dy}{dx} + y\left(\frac{1}{x^2}\right) = 0$; $P = \frac{1}{x^2}$, $Q = 0$

2. A differential equation is of the form $\frac{dx}{dy} + Px = Q$, where P & Q are functions of y or constants is known as a first order differential equation.

$$\frac{dx}{dy} + Px = Q$$

P & Q are constants or functions in variable y only.

eg. $\frac{dx}{dy} + (\sin y)x = \cos^2 y$

$P = \sin y$, $Q = \cos^2 y$

Steps involved to solve First Order Linear Differential Equation

1. Write the given differential equation in the form $\frac{dy}{dx} + Py = Q$. & identify P & Q

2. Find integrating factor $(I.F) = e^{\int P dx}$

3. $y \times I.F = \int (Q \times IF)\, dx + c$

$$\frac{dx}{dy} + Px = Q$$

① Identify P & Q

② $I.F = e^{\int P dy}$

③ $x \times IF = \int (Q \times IF)\, dy + c$

$e^{\log f(x)} = f(x)$

$e^{\log \sin x} = \sin x$

$e^{\log \sec x} = \sec x$.